Minimum Variance Portfolio

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Introduction

This project tests a risk-minimizing investment strategy using a dynamic Minimum Variance Portfolio (MVP) based on daily return data. The portfolio includes:

- Pfizer (PFE), a pharmaceutical company.
- Rheinmetall (RHM), a German defense and automotive firm.

Portfolio risk is measured over time using Value-at-Risk (VaR) and Expected Shortfall (ES), especially during times when the market is stressed.

Objectives

- Create a daily-updated MVP using recent return data.
- Measure risk with different VaR and ES methods.
- Test performance in stable and volatile periods (e.g., COVID-19, Ukraine war).
- Improve risk estimates using deep learning and stress scenarios.

Data Collection

Daily stock prices for Pfizer and Rheinmetall were collected from **January 1, 2010, to December 31, 2025** using the `yfinance` Python library.

Observations

- **Pfizer (PFE):** Prices changed during the COVID-19 pandemic (2020–2021), mostly because of vaccine news and health rules.

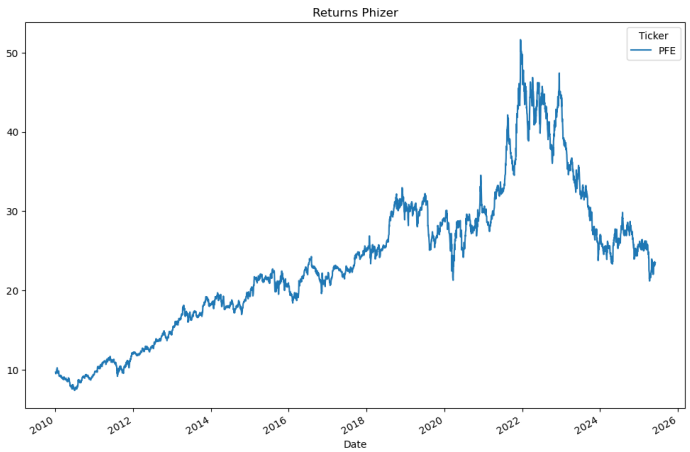


Figure 1: Pfizer Stock Price

- **Rheinmetall (RHM):** Big price increases from 2022 to 2025, mainly due to the Ukraine war and more defense spending in Europe.

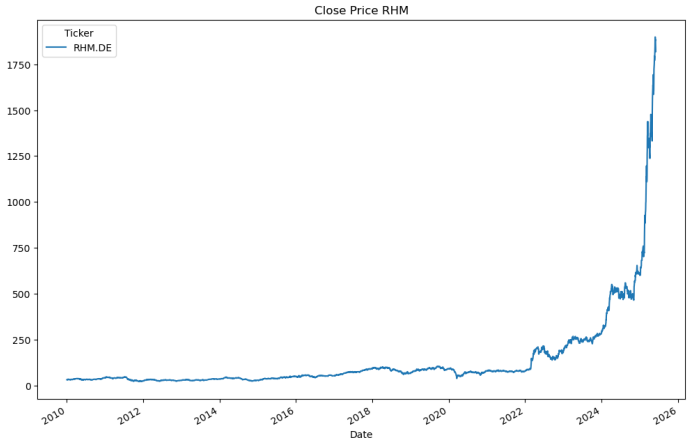


Figure 2: Rheinmetall Stock Price

Data Preparation

Calcolationg the daily returns for Pfizer and Rheinmetall were calculated from stock prices using the standard formula:

$$\text{Return}_t = \frac{\text{Price}_t - \text{Price}_{t-1}}{\text{Price}_{t-1}}$$

The return series were then merged into one dataset, and missing values (NaNs) were removed for clean analysis.

Daily Returns Observation

- **Pfizer:** Highly volatile during COVID-19, reacting strongly to pandemic-related news.

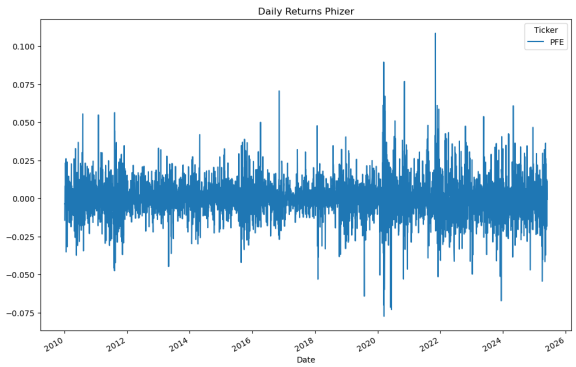


Figure 3: Pfizer Daily Returns

- **Rheinmetall:** Generally stable, with occasional spikes due to geopolitical tensions.

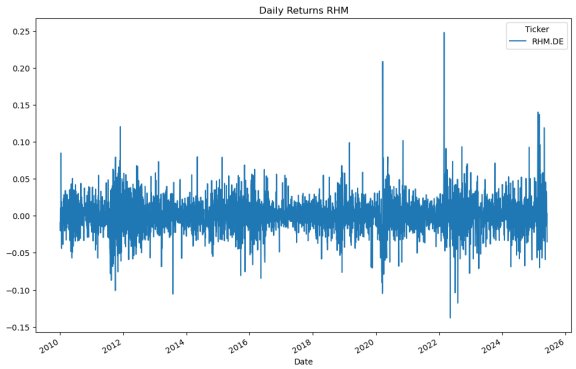


Figure 4: Rheinmetall Daily Returns

Portfolio Construction

The data is splited into two parts: a training set from 2010 to 2023 used to build the model, and a testing set from 2024 to 2025 used to evaluate its performance.

Dynamic Minimum Variance Portfolio

The covariance matrix was calculated daily using the past 20 days of returns. This matrix was then used to calculate the portfolio weights that minimize variance. The weight for Pfizer is calculated as:

$$w_{\text{PFE}} = \frac{\text{Var}_{\text{RHM}} - \text{Cov}_{\text{PFE,RHM}}}{\text{Var}_{\text{PFE}} + \text{Var}_{\text{RHM}} - 2 \cdot \text{Cov}_{\text{PFE,RHM}}}$$

The weight for Rheinmetall is:

$$w_{\text{RHM}} = 1 - w_{\text{PFE}}$$

Weights are updated every day to follow the latest market changes. This helps the portfolio adjust based on how prices move and how the stocks relate to each other, keeping the risk as low as possible.

Weights Behavior

The portfolio changed its investments over time to match the market. During the COVID-19 pandemic, it invested more in Pfizer because the company was more stable and related to vaccine news. In 2022, the portfolio moved more money into Rheinmetall because of higher values in defense due to world events. Also, short-selling is shown, which means the portfolio can make money when a stock's price goes down. This helps lower risk by balancing out losses from other stocks.

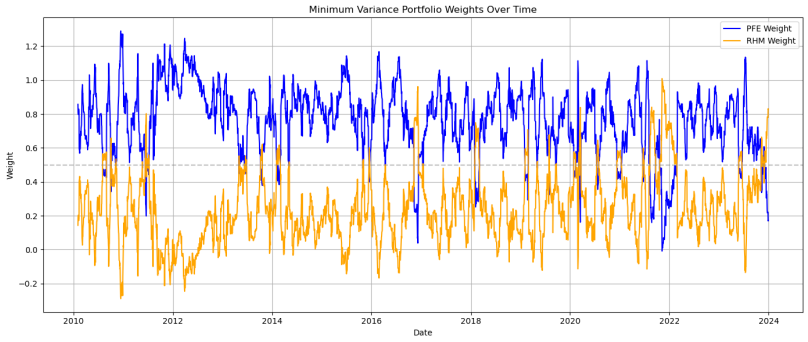


Figure 5: Daily Portfolio Weights

Portfolio Value and Quantities

The portfolio starts with 10,000 units of money. Every day, the portfolio return is calculated by adding the returns of Pfizer and Rheinmetall stocks, each multiplied by how much of the portfolio is invested in them. This is shown by the formula:

$$r_{\text{portfolio},t} = w_{\text{PFE},t} \times r_{\text{PFE},t} + w_{\text{RHM},t} \times r_{\text{RHM},t}$$

Using this daily return, the portfolio value at day t is found by multiplying the value from the previous day by $1 + r_{\text{portfolio},t}$:

$$V_t = V_{t-1} \times (1 + r_{\text{portfolio},t})$$

Figure 6 shows how the portfolio value changes over time. It generally grows, but there is a drop in 2020 because of the COVID-19 pandemic when stock prices went down.

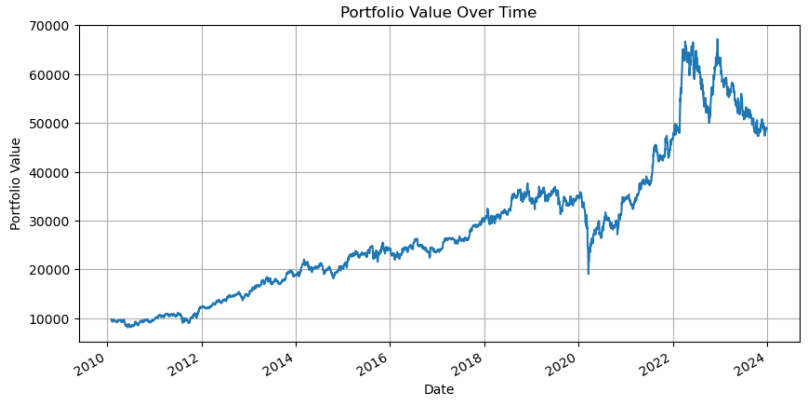


Figure 6: Portfolio Value Over Time

Next, to find out how many shares of each stock are held, the money invested in each stock (which is the total portfolio value multiplied by its weight) is divided by that stock's price on day t:

$$q_{\text{PFE},t} = \frac{V_t \times w_{\text{PFE},t}}{\text{Price}_{\text{PFE},t}}, \quad q_{\text{RHM},t} = \frac{V_t \times w_{\text{RHM},t}}{\text{Price}_{\text{RHM},t}}$$

Because Pfizer's price is usually lower than Rheinmetall's, the portfolio holds more shares of Pfizer. This is why the number of Pfizer shares is higher, as shown in Figure 7. The number of shares changes every day depending on the portfolio value, weights, and stock prices.

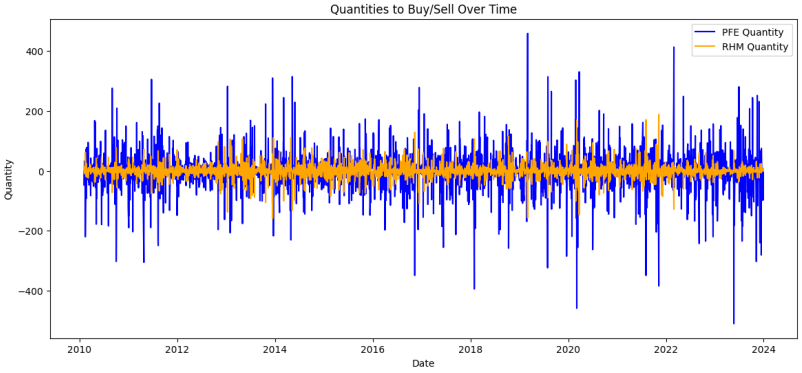


Figure 7: Number of Shares Held Over Time

Risk Measurement

Risk is measured using four different methods, all based on the training data (2010–2024). A grid search approach was applied to find the best parameters, such as the size of the rolling window and the thresholds, for each method to improve precision.

The methods used are:

1. Historical VaR/ES **2. Z-score Method**

3. LSTM Forecasting **4. Stress Amplification**

Historical Method

The Historical method calculates Value-at-Risk (VaR) and Expected Shortfall (ES) by using the worst 5% of portfolio returns over a rolling window. Applying grid search on the training data identified a 60-day window as the best choice. A larger window captures more data, resulting more stable in risk examination.

Results on Training Set:

- VaR Violation Rate: 6.15% — a bit higher than the expected 5%, meaning the model is mostly accurate but slightly too low in predicting risk.
- Expected Shortfall (ES): 0.0235 — the average loss during the worst 5% of cases, showing how big losses usually are in bad times.

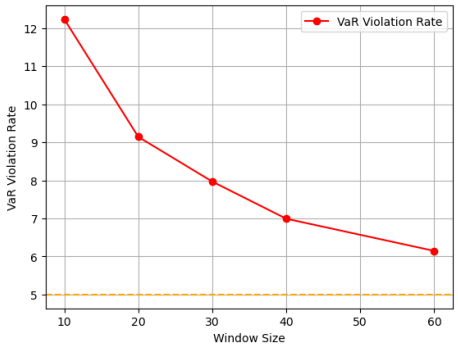


Figure 8: Historical Grid

Z-score Method

The Z-score method assumes that returns follow a normal distribution. It calculates VaR and ES using the average return μ and standard deviation σ of returns:

$$\text{VaR}_\alpha = \mu - z_\alpha \cdot \sigma, \quad \text{ES}_\alpha = \mu - \frac{\sigma \cdot \phi(z_\alpha)}{1-\alpha}$$

Here, z_α is the standard normal quantile, and ϕ is the normal density function.

Using grid search on the training data to find the best parameters:

```
{'window': 60, 'stress_multiplier': 3.0, 'z_threshold': 3.0}
```

The 60-day window captures more data, resulting more stable in risk examination. The stress multiplier and z-threshold are both set to 3.0, the highest values tested. This means the model increases risk only during strong market stress, so it doesn't overreact to small changes and focuses on real dangers.

Results on Training Set:

- VaR Violation Rate: 6.15% — slightly above the expected 5%, meaning the model is mostly accurate but a little too low in predicting risk.
- Expected Shortfall (ES): 0.0235 — the average loss during the worst 5% of cases, showing typical size of large losses.

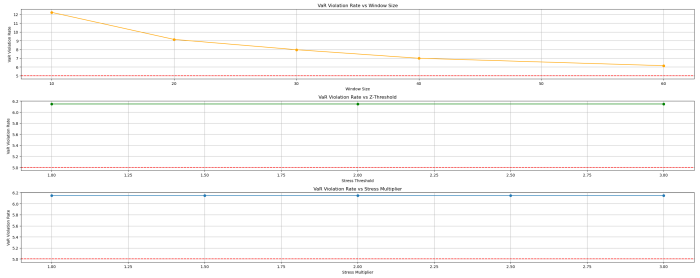


Figure 9: Z-score Grid

Stress Amplification

This method raises risk estimates when the market shows signs of stress. It uses a 60-day rolling window and increases risk by a factor when volatility goes above a set limit. This helps the model better detect risk during unstable times.

The best settings found with grid search are:

```
{'window': 60, 'stress_threshold': 0.01, 'stress_multiplier': 3.0}
```

The 60-day window balances reacting to recent changes and keeping results steady. The stress multiplier of 3.0 increases risk estimates when volatility is higher than 1%, making the model more careful in risky situations.

Results on Training Set:

- VaR Violation Rate: 2.78% — lower than the expected 5%, showing the model is careful and does not miss many risks.
- Expected Shortfall (ES): 0.0205 — the average loss during the worst 5% of cases, showing typical big losses.

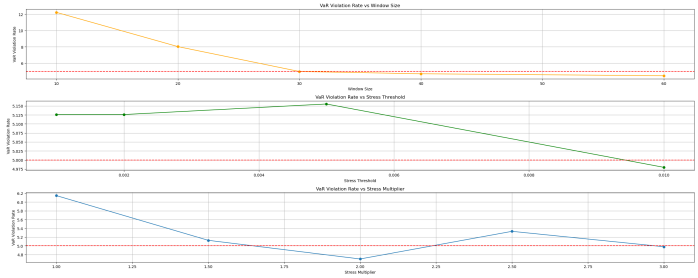


Figure 10: Stress Amplification Grid

LSTM Method

The LSTM method uses a deep learning model to predict future returns based on past data. It learns complex patterns and can adjust quickly to market changes.

Results on Training Set:

- VaR Violation Rate: 5.16% — very close to the expected 5%, showing good accuracy.
- Actual ES: 0.0251 — the average loss during the worst 5% of cases.

Evaluation

Models were evaluated on the test set to assess their performance based on the best parameters that was found on the trainig set.

Historical Method

This method estimates risk using the worst 5% of past returns. On the test set, it showed a slightly higher VaR violation rate than expected, suggesting some underestimation of risk. The Expected Shortfall reflects typical large losses.

Results on Test Set:

- VaR Violation Rate: 7.29% — slightly above 5%, indicating some missed risk.
- Expected Shortfall (ES): 0.0266 — average loss in the worst 5% cases.

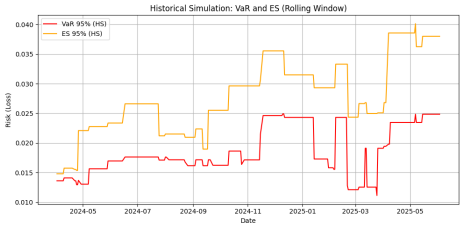


Figure 11: Historical Method Results on Test Set

Stress Amplification Method

This method raises risk estimates during high volatility. On the test set, it had a very low VaR violation rate, showing a cautious approach, and a higher Expected Shortfall reflecting stress conditions.

Results on Test Set:

- VaR Violation Rate: 0.69% — well below 5%, indicating strong caution.
- Expected Shortfall (ES): 0.0338 — larger average loss due to stress adjustment.

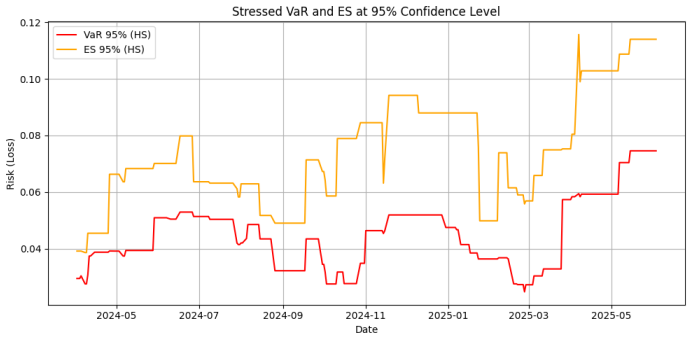


Figure 12: Stress Amplification Method Results on Test Set

Stress Amplification with Z-Score Method

This method filters for extreme volatility before adjusting risk. Its test results closely match the Historical method, with a somewhat high VaR violation rate and similar Expected Shortfall.

Results on Test Set:

- VaR Violation Rate: 7.29% — slightly above 5%, similar to Historical.
- Expected Shortfall (ES): 0.0266 — comparable average loss.

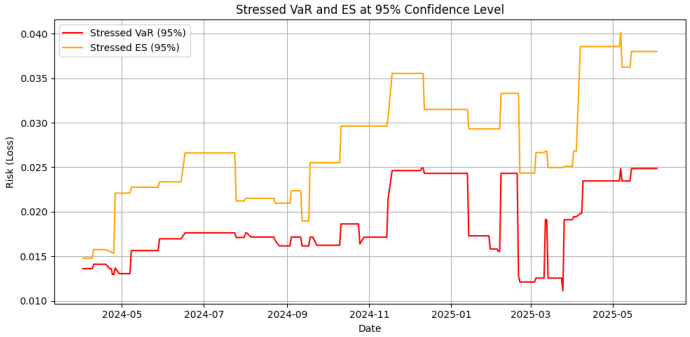


Figure 13: Stress Amplification with Z-Score Method Results on Test Set

LSTM Method

The LSTM adapts quickly to market changes and predicted returns using deep learning. On the test set, it achieved a VaR violation rate very close to 5%, showing reliable accuracy, with a moderate Expected Shortfall.

Results on Test Set:

- VaR Violation Rate: 4.70% — close to the expected 5%, indicating good performance.
- Expected Shortfall (ES): 0.0296 — average loss in the worst 5% of cases.

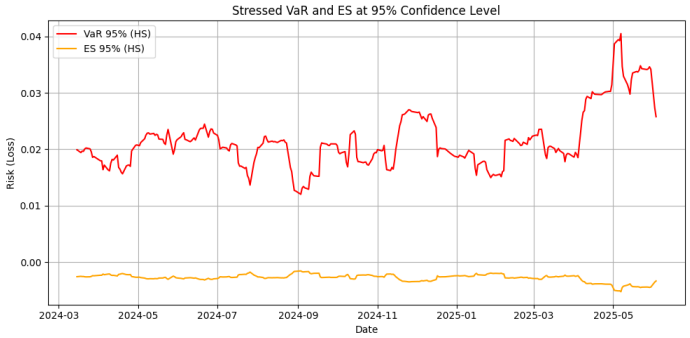


Figure 14: LSTM Method Results on Test Set

Models Comparison

Among all tested methods, the LSTM model performed the best. It showed the closest VaR violation rate to the expected 5%, indicating more accurate risk estimation. Its ability to learn complex patterns and adapt quickly to market changes likely contributed to its superior forecasting performance. The other models, while useful, either underestimated or overestimated risk, especially during volatile periods.

Conclusion

This project tested different ways to measure risk in a portfolio with Pfizer and Rheinmetall stocks. The LSTM method worked best because it predicted risk closest to the expected 5% rate. Traditional methods like Historical and Stress Amplification were less accurate when markets changed quickly. The results show that deep learning approaches can improve risk measurement and help investors make better decisions, especially during volatile market periods.

Bibliography

- Yahoo Finance, Pfizer Inc. (PFE) Stock Data: `https://finance.yahoo.com/quote/PFE/`
- Yahoo Finance, Rheinmetall AG (RHM) Stock Data: `https://finance.yahoo.com/quote/RHM.DE/`